

Yunhan Zheng

55 Normanton Park, Singapore 117283

+65 98728116 | yunhan@mit.edu

EDUCATION

- Sep 2021 - Feb 2024 **Massachusetts Institute of Technology (MIT), USA**
PhD, Interdepartmental Program in Transportation
- *Advisor:* Professor Jinhua Zhao
 - *Research Topic:* Assessing the Environmental, Economic, and Social Impacts of Sustainable Mobility Solutions
- Sep 2018 - Jul 2021 **Massachusetts Institute of Technology (MIT), USA**
Master of City Planning & Master of Science in Transportation
- *Advisor:* Professor Jinhua Zhao
 - *Research Topic* Computational Fairness in Travel Demand Modeling
 - *GPA:* 4.9/5.0
- Sep 2014 - Jul 2018 **Peking University (PKU), China**
Bachelor of Urban Management, School of Government, *GPA:* 3.93/4.0
Bachelor of Economics, National School of Development, *GPA:* 3.87/4.0

PROFESSIONAL EXPERIENCE

- Feb 2024-present **Singapore-MIT Alliance for Research and Technology, Singapore**
Postdoc Associate
⇒ *Project:* Integrated Human-Machine Intelligence for Economic Growth | AI for Human Capital Development in Future of Work
- 2018-2023 **Massachusetts Institute of Technology, Cambridge, MA, USA**
Graduate Research Assistant at MIT Transit Lab
- *Collaboration Agencies:* Singapore-MIT Alliance for Research and Technology | World Resources Institute (WRI) | MIT Energy Initiative | U.S. Energy Foundation | The Barr Foundation

RESEARCH INTERESTS

Research Areas: Urban and Transportation Economics, Urban Computing and Machine Learning, Travel Decisions and Urban Network, Transportation Policies, Intelligent Transportation System

Research Vision: My research area is the intersection of Causal Inference, Urban Computing, and Travel Decisions: (1) Causal Inference, Transportation Economics and Policy: Leveraging advanced causal inference methods for the assessment of emerging transportation tools like congestion pricing, EV charging infrastructure, and teleworking on mobility, environment, economy, and social equity. (2) Urban Computing and Machine

Learning: Developing machine learning algorithms to predict and nudge travel behaviors; developing bias-mitigation methods to enhance fairness and equity in machine learning algorithms. (3) Travel Decisions and Urban Network: Combining statistical modeling, deep learning computation, and network theory for the modeling and analysis of individual travel decisions and urban networks.

PUBLICATIONS

(* indicates corresponding author; † indicates co-first authors)

Selected Publications

- 2025 **Robust discrete choice model for travel behavior prediction with data uncertainties**
Baichuan Mo, Yunhan Zheng*, Xiaotong Guo, Ruoyun Ma, Jinhua Zhao
European Journal of Operational Research
- 2025 **Impacts of the built environment on remote work choices and preferences**
Meilin Yang, Yunhan Zheng*, Shenhao Wang, Xiaotong Guo, Jinhua Zhao
Cities, 163, 106034. (SSCI, IF: 6.0)
- 2024 **Effects of electric vehicle charging stations on the economic vitality of local businesses**
Yunhan Zheng*, David R. Keith, Shenhao Wang, Mi Diao, Jinhua Zhao
Nature Communications, 15, 7437.
⇒ Covered by 30 news outlets, including *the Washington Post*, *CBC Canada*, *Lianhe Zaobao*, *MIT News*
⇒ Selected as *Editors' Highlights* by *Nature Communications*
⇒ Ranked in the top 1% for online attention among articles of similar age across all journals.
- 2024 **Impacts of remote work on vehicle miles traveled and transit ridership in the United States**
Yunhan Zheng, Shenhao Wang*, Lun Liu, Jim Aloisi, Jinhua Zhao
Nature Cities, 1-13.
⇒ Covered by 76 news outlets, including *Tech Times*, *MIT News*
⇒ Ranked in the top 1% for online attention among articles of similar age across all journals.
- 2024 **Infrequent activities predict economic outcomes in major American cities**
Shenhao Wang†, Yunhan Zheng†, Guang Wang, Takahiro Yabe, Esteban Moro, Alex 'Sandy' Pentland
Nature Cities, 1(4), 305-314. († indicates co-first authors)
- 2023 **Impacts of congestion pricing on ride-hailing ridership: evidence from Chicago**
Yunhan Zheng*, Patrick Meredith-Karam, Anson Stewart, Hui Kong, Jinhua Zhao.
Transportation Research Part A: Policy and Practice, 170, 103639. (SCI/SSCI, IF: 6.3)
- 2023 **Examining the interactions between working from home, travel behavior and change in car ownership due to the impact of COVID-19**
Yunhan Zheng*, Nicholas S Caros, Jim Aloisi, Jinhua Zhao.
Travel Behaviour and Society, 33, 100634. (SSCI, IF: 5.1)
- 2023 **Fairness-enhancing deep learning for ride-hailing demand prediction**

Yunhan Zheng, Qingyi Wang, Dingyi Zhuang, Shenhao Wang*, Jinhua Zhao
IEEE Open Journal of Intelligent Transportation Systems, vol. 4, pp. 551-569. (ESCI, IF: 4.6)

- 2022 **Gender differences in the user satisfaction and service quality improvement priority of public transit bus system in Porto Alegre, Brazil**
Yunhan Zheng, Hui Kong*, Guillermo Petzhold, Mariana M. Barcelos, Christopher P. Zegras, Jinhua Zhao.
Travel Behaviour and Society, 28, 22-37. (SSCI, IF: 5.1)
- 2021 **Equality of opportunity in travel behavior prediction with deep neural networks and discrete choice models**
Yunhan Zheng, Shenhao Wang*, Jinhua Zhao
Transportation Research Part C: Emerging Technologies, 132, 103410. (SCI, IF: 7.6)
- 2021 **Measuring policy leakage of Beijing's car ownership restriction**
Yunhan Zheng, Joanna Moody, Shenhao Wang, Jinhua Zhao*
Transportation Research Part A: Policy and Practice, 148, 223-236 (SCI/SSCI, IF: 6.3)
- 2021 **User satisfaction and service quality improvement priority of bus rapid transit in Belo Horizonte, Brazil**
Yunhan Zheng, Hui Kong*, Guillermo Petzhold, Mariana M. Barcelos, Christopher P. Zegras, Jinhua Zhao.
Case Studies on Transport Policy, 9(4), 1900-1911.

Other Publications

- 2025 **GETS: Ensemble Temperature Scaling for Calibration in Graph Neural Networks**
Dingyi Zhuang, Chonghe Jiang, Yunhan Zheng, Shenhao Wang, Jinhua Zhao
The Thirteenth International Conference on Learning Representations (ICLR)
- 2024 **Comparing hundreds of machine learning and discrete choice models for travel demand modeling: An empirical benchmark**
Shenhao Wang, Baichuan Mo, Yunhan Zheng, Stephane Hess, Jinhua Zhao
Transportation Research Part B: Methodological, 179, 102869. (SCI, IF: 5.8)
- 2024 **Deep hybrid model with satellite imagery: how to combine demand modeling and computer vision for behavior analysis?**
Qingyi Wang, Shenhao Wang*, Yunhan Zheng, Hongzhou Lin, Xiaohu Zhang, Jinhua Zhao, Joan Walker
Transportation Research Part B: Methodological, 179, 102869. (SCI, IF: 5.8)
- 2024 **Simulating Integration of Urban Air Mobility into Existing Transportation Systems: Survey**
Xuan Jiang, Yuhan Tang, ..., Yunhan Zheng, Jinhua Zhao, Raja Sengupta
Journal of Air Transportation, 32(3), 97-107.
- 2021 **User satisfaction and service quality improvement priority of bus rapid transit in Belo Horizonte, Brazil**
Yunhan Zheng, Hui Kong*, Guillermo Petzhold, Mariana M. Barcelos, Christopher P. Zegras, Jinhua Zhao.
Case Studies on Transport Policy, 9(4), 1900-1911.
- 2021 **Dispersion of agglomeration through high-speed rail in China**

Wanli Fang, Yunhan Zheng, Mi Diao, Jinhua Zhao.
In *Urban Form and Accessibility* (pp. 327-357). Elsevier.

Forthcoming

- Under Review **Consumption Complexity as a Driver of Urban Economic Development: A Nationwide Analysis of the United States**
Yunhan Zheng, Yuebing Liang, Donghang Li, Dingyi Zhuang, Shenhao Wang, Jinhua Zhao
- Under Review **Causal effects of job-housing imbalance on remote work**
Donhang Li, Yunhan Zheng*, Shenhao Wang, Xiaotong Guo, Jinhua Zhao
- Working **Fairness-Enhancing Vehicle Rebalancing in the Ride-hailing System**
Xiaotong Guo, Hanyong Xu, Dingyi Zhuang, Yunhan Zheng*, Jinhua Zhao

INVITED TALKS & GUEST LECTURES

- Jan 2025 **Urban Transportation Governance in the Era of Big Data and Artificial Intelligence**
School of Public Policy & Management, Tsinghua University
- Dec 2024 **Sustainable Mobility Solutions and Evolving Travel Patterns in the New Era**
At the 6th PKU-CUES International Young Scholar Forum, Peking University
- Nov 2024 **Special Lecture on the Development Ideas of Charging Infrastructure during the 15th Five-Year Plan Period**
Energy Research Institute of the National Development and Reform Commission (NDRC)
- Jul 2024 **Urban Activities and Transportation Behavior in the New Era: Interconnections and Impact Mechanisms**
School of Architecture and Civil Engineering, Xiamen University
- Jan 2024 **Measuring, Evaluating, and Enhancing Algorithmic Fairness in Traffic Demand Prediction and Traffic Service Provision**
At the 6th Peking University Intelligent Discipline Youth Forum, Peking University
- May 2023 **Algorithmic Fairness in Travel Demand Prediction**
At MIT Mobility Initiative Forum
- Apr 2023 **Algorithmic Fairness in Travel Demand Prediction**
At Urban Artificial Intelligence Laboratory, University of Florida
- Apr 2021 **Measuring policy leakage of Beijing's car ownership restriction**
In “11.478 Behavioral Science and Urban Mobility” class, MIT

TEACHING EXPERIENCE

- Fall 2023 **Class “Behavioral Science, Artificial Intelligence, and Urban Mobility”**
Teaching assistant, at MIT Department of Urban Studies and Planning
- Fall 2017 **Class “Economic Geography”**
Teaching assistant, at School of Government of Peking University

CONFERENCE PRESENTATIONS

- Jan 2025 **Effects of electric vehicle charging stations on the economic vitality of local businesses**
Transportation Research Board (TRB) 104th Annual Conference, Washington, D.C., USA
(Guest speaker at the Committee on Economic Development and Land Use meeting)
- Nov 2024 **Impacts of remote work on vehicle miles traveled and transit ridership in the United States**
ACSP 2024 Annual Conference, Seattle, USA
- Nov 2022 **Fairness-enhancing spatiotemporal demand prediction for ride-hailing services**
ACSP 2022 Annual Conference, Toronto, Canada
- Jan 2021 **Race, gender, and income disparity in travel behavior prediction with machine learning**
Transportation Research Board (TRB) 100th Annual Conference, Washington, D.C., USA
- Jan 2021 **User satisfaction and service quality improvement priority of bus rapid transit in Belo Horizonte, Brazil**
Transportation Research Board (TRB) 100th Annual Conference, Washington, D.C., USA
- Jan 2020 **Measuring policy leakage of Beijing's car ownership restriction in neighboring cities**
Transportation Research Board (TRB) 99th Annual Conference, Washington, D.C., USA
- Oct 2019 **Nudging price information in China's residential real estate market**
ACSP 2019 59th Annual Conference, Greenville, USA

SELECTED AWARDS AND HONORS

- 2018 - 2019 **DUSP Graduate Student Fellowship**, MIT
- 2018 **Outstanding Graduates**, Peking University
Top 10% of all graduates at PKU
- 2017 **National Scholarship**, Peking University
Top 3% of all undergraduate students at PKU
- 2017 **Silver Medalist**, National Competition of City Governance, Shanghai, China
Team leader, Silver Medalist out of 54 teams
- 2015 & 2016 **Leo KoGuan Scholarship**, Peking University
Top 10% of all undergraduates

ACADEMIC SERVICE

- Journal Referees Scientific Data | Transportation Research Part A | Travel Behaviour and Society | Transportation | Journal of Public Transportation | Journal of Advanced Transportation
- Conference Referees Transportation Research Board Annual Meeting (TRB)

SKILLS

Programming	Python, R, Julia, ArcGIS, QGIS, SQL, JavaScript, STATA, MS Office
Machine Learning Framework	Pytorch, TensorFlow
Analytical Skills	Econometrics, Statistics, Deep Learning, Computer Vision, Spatial Visualization and Data Management, Discrete Choice Analysis, Web Scrapping, Hypothesis Testing, Transportation Policy Analysis